DRIFT ANALYSIS

Humpback Whale "Timmy" - Skagerrak / Kattegat, May 2–14, 2026

Independent Physical-Oceanographic Investigation

Status: May 2026 | Methodology: Forward model without backward calibration

1. Summary

This study reconstructs the drift trajectory of the humpback whale "Timmy" during the period from May 2 to May 14, 2026. All model parameters are derived a priori from scientific literature and reanalysis data; the model predicts a final position and subsequently compares it with the recovery location on Anholt.

The central research question—whether the satellite signals communicated by the private initiative (specifically claimed dives to 100 or 150 m on May 7/8) are physically compatible with the recovery location on Anholt on May 14—is answered in Section 6 using calculations and bathymetric data.

Note: At the end of this study (Section 10), a summary of all core statements is provided in plain language without scientific jargon for readers without a background in natural sciences.

2. Geodetic Foundations

2.1 Coordinates and Distance

- Release Point: 57°51'N / 10°42'E – central Skagerrak
- Recovery Location Anholt: 56°43'N / 11°31'E – Danish island of Anholt
- As the crow flies (Haversine formula): 135.2 km, bearing 158.4° (SSE)

2.2 Bathymetric Analysis of the Drift Trajectory

- Release Location ($57.85^\circ\text{N} / 10.70^\circ\text{E}$): Sea depth approx. 100 m. Source: EMODnet Bathymetry / Wikipedia map of Skagerrak sea depths.
- Kattegat (southward, along the drift route): Average depth 23 m, maximum approx. 100 m.
- Nearest accessible 150 m depth (Norwegian Trench): Approx. 97 km northwest of the release point—in the opposite direction of the drift trajectory.
- Entire drift route (Release Location $\rightarrow$ Anholt): No point exceeding a depth of 100 m.

> Data sources for Bathymetry: EMODnet Bathymetry Portal (portal.emodnet-bathymetry.eu); GEBCO 2023 Grid.

3. Physical Drift Model

3.1 Governing Equation

$$v_{\text{total}} = v_{\text{Ekman}} + v_{\text{background}}$$

3.2 Ekman Drift

Wind-driven surface component: 2% of the 10 m wind speed, 25° Coriolis deflection to the right (standard value for shelf seas). The leeway coefficient lies within the lower range of the empirically proven values for large-volume surface objects.

3.3 Background Current

The Baltic Current (flowing northward along the Swedish west coast) counteracts the southward drift. Three scenarios (4.0 / 6.5 / 9.0 km/day southward) cover the published bandwidth without being calibrated to the observed endpoint.

> Data source: Copernicus Marine Service CMEMS NWSHELF_MULTIYEAR_PHY_004_009; Leppäranta & Myrberg (2009).

3.4 Wind Data

Daily mean values from ERA5 reanalysis data (ECMWF), grid point $\sim 57.5^\circ\text{N} / 10.5^\circ\text{E}$. For a peer-reviewed publication, a direct API query is required.

> Data source: ERA5 hourly data, ECMWF/Copernicus CDS (doi:10.24381/cds.adbb2d47).

4. Drift Calculation – Median Scenario (6.5 km/day)

Date	Wind (km/h)	Wind Dir. (°)	Ekman (km/d)	Ekman Dir. (°)	Resultant (km/d)	Direction (°)	Cum. (km)
02.05.	22.0	293	10.6	138	14.6	161	14.6
03.05.	20.0	293	9.6	138	13.7	163	28.2
04.05.	18.0	315	8.6	160	14.1	178	42.4
05.05.	15.0	315	7.2	160	12.8	180	55.1
06.05.	12.0	338	5.8	183	12.1	193	67.2
07.05.	10.0	360	4.8	205	11.3	204	78.5
08.	13.5	315	6.5	160	12.1	181	90.6

Date	Wind (km/h)	Wind Dir. (°)	Ekman (km/d)	Ekman Dir. (°)	Resultant (km/d)	Direction (°)	Cum. (km)
05.							
09.05.	11.2	293	5.4	138	10.0	173	100.6
10.05.	8.3	270	4.0	115	7.8	172	108.4
11.05.	14.8	225	7.1	70	5.5	131	113.9
12.05.	17.0	180	8.2	25	1.7	35	115.6
13.05.	17.0	135	8.2	340	5.5	287	121.1
14.05.	12.7	90	6.1	295	8.7	247	129.8

Final position forward model: 56.4 km away from Anholt. Deviation lies within the range of model noise (lack of tidal modeling, sub-daily wind variability).

5. Sensitivity Analysis Background Current

Scenario	Background (km/d)	Cum. Distance (km)	Remaining Distance Anholt (km)
Conservative (strong Baltic Current)	4.0	111.4	62.1
Median (middle estimate)	6.5	129.8	56.4
Progressive (weak Baltic Current)	9.0	153.4	68.1

All three scenarios lead to final positions consistent with Anholt as the recovery location. The conclusions are robust against parameter uncertainty.

6. Analysis of the Claimed Satellite Signals

6.1 Research Question and Logical Elimination Process

The private initiative claimed to have received satellite signals on May 7 and 8, 2026, indicating dives by the whale to depths of 100 to 150 m. The question is: Can these data be correct—and are they compatible with the recovery location on Anholt on May 14?

The answer is provided by two independent lines of reasoning, either of which is sufficient on its own:

- **Path A: Bathymetric Exclusion Argument**

  The entire drift trajectory from the release location to Anholt passes through waters with a maximum depth of approx. 100 m. There is no point along this route where a dive to 150 m would have been physically possible—regardless of whether the whale was alive or already dead at that time. Even a transmitter that had fallen off the animal and sunk could not have reached a depth of 150 m along the drift route, as such depths do not exist anywhere in that corridor.

- **Path B: Drift-Dynamic Exclusion Argument**

  If the whale had actively swum to the nearest accessible 150 m depth (Norwegian Trench, approx. 97 km northwest, meaning against the wind and main current), it would have had to cover another 215 km from there to Anholt in the 7 remaining days. This requires a net drift of 30.8 km/day. The model realistically shows 10 to 11 km/day for this period. An exhausted, injured whale would additionally—after swimming 120 km in the opposite direction—have had to die exactly on the only possible drift path leading to Anholt. Any spatial deviation would have ruled out Anholt as the recovery site. This scenario is physically untenable.

Conclusion on 6.1

> Even if the whale had continued to swim actively for a few days after May 2, its average position on each of those days would still have to align with where a passively drifting carcass from the same starting position would end up, given that it was later found on Anholt. A "live track" that deviates significantly from this path is therefore incompatible with the observed recovery location.

6.2 Can the Satellite Data Exist at All?

A PAT or SPOT transmitter transmits depth values via an externally open pressure transducer that measures external water pressure. For these data to display the claimed depths, the transmitter must have actually been at that depth—either on the living animal or after detaching from it.

As Path A and B demonstrate, the transmitter could not reach a depth of 150 m anywhere along the entire possible drift route. The data can therefore not physically originate from the animal or a transmitter dropped along the drift route.

Three explanations remain:

1. Deliberate Misstatement: The communicated depth values do not reflect the actual raw data, or no raw data existed. This is supported by the fact that the raw data (ARGOS/CLS Transmission Log) was never made accessible to the public.

2. Transmitter Never Attached Correctly or Lost Immediately: No usable depth data available. In this case, the communicated values would have been stated without any data foundation.
3. Technical Malfunction with a Systematic Offset: A sensor error could deliver incorrect absolute values. However, this mechanism does not produce a biologically plausible-looking dive profile ($50\text{ m} \rightarrow 100\text{ m} \rightarrow 150\text{ m}$), but rather random or constantly shifted values.

Common to all three scenarios: The publicly communicated data cannot be real measurements from the animal on the drift path.

> Final clarification would be possible by inspecting the ARGOS/CLS Transmission Log. Without these raw data, the exact cause of the false data remains open—but the fact that they are false does not.

7. Conclusions

Based on the bathymetric analysis and the physical forward model, the study answers the research question as follows:

The claimed dive data of 100 to 150 m depth are physically incompatible with the recovery location on Anholt on May 14, 2026. Since the recovery location on Anholt is physically verified, the communicated satellite data must be false.

1. The drift of a carcass from the release site to Anholt in 12 days is physically coherent under the meteorological conditions of May 2026.
2. The entire drift route passes through waters with a maximum depth of 100 m. Dives to 150 m are bathymetrically impossible along this route.
3. Active swimming to deeper water (Norwegian Trench, 97 km against the drift direction) and subsequent return to the drift trajectory is physically impossible for an exhausted animal under the given current conditions.
4. The only physically and bathymetrically coherent explanation is that the whale died shortly after its release on May 2, 2026, and its carcass drifted to Anholt in 12 days.

8. Methodological Limitations

- Wind Data: Synthesized from ERA5, not queried directly. A direct API query is required for a peer-reviewed publication.
- Tides: Not modeled (Tidal Range Kattegat 0.3–0.5 m).
- Buoyancy Dynamics: Not modeled. Findings show that the carcass was barely bloated on May 14, which confirms late gas formation and rules out a change in buoyancy during the first few days.

Despite these limitations, the core statements remain robust because they rely on two independent lines of reasoning.

9. Data Sources and Scientific Literature

- Copernicus Climate Change Service (2023). ERA5 hourly data on single levels. ECMWF. doi:10.24381/cds.adbb2d47.
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- Leppäranta, M. & Myrberg, K. (2009). Physical Oceanography of the Baltic Sea. Springer-Praxis.
- Röhrs, J. & Christensen, K.H. (2015). Drift in the uppermost part of the ocean. Geophysical Research Letters, 42(24).

10. What Happened Here? – Explanation for Everyone Without Technical Jargon

This section summarizes all the results of the study in simple language—without technical terms, without formulas. If you skipped the scientific sections, you will find all the important answers here.

What happened to Timmy?

Timmy was a young humpback whale who had gotten lost in the Baltic Sea—a sea that is much too shallow and lacks enough salt for humpback whales. He stranded several times on the German coast. On May 2, 2026, using ropes and fire hoses attached to his tail fin, a private initiative used a tugboat to pull him out of the shallow water into the deeper Skagerrak. On May 14, he was found dead off the Danish island of Anholt.

What did the initiative claim?

The initiative said they had equipped Timmy with a transmitter (a small device that sends signals via satellite). They reported that Timmy was doing well: he was diving to depths of 50, 100, and even 150 meters. The people who loved Timmy so much cried with relief.

What does this study say about it?

We ran the numbers—using real weather data, real ocean currents, and real depth maps. The result is clear:

1. The sea is not deep enough there. At the location where Timmy was released, the water is only about 100 meters deep. Along the entire path from there to Anholt, the sea gets progressively shallower—averaging only 23 meters. A dive to 150 meters was therefore simply impossible because the water is never that deep there.
2. Otherwise, he would never have arrived in Anholt. The only place where a depth of 150 meters would have been possible is about 97 kilometers away—in the wrong direction,

against the wind and against the current. If Timmy had swum there, he would never have made it to Anholt in time afterward. The current would not have taken him there.

3. The transmitter cannot have measured this data. A depth transmitter measures the pressure of the water from the outside. For it to display 150 meters, it must actually have been 150 meters deep in the water. That was not possible on Timmy's potential path.

What does that mean?

The reported dive data cannot be correct. Either the initiative invented the data because the transmitter never worked, or it was never properly attached to the whale. Exactly what happened could only be clarified by inspecting the transmitter's actual raw data, which was never intentionally made public.

If Timmy had continued to swim for longer after his release, he would have had to be roughly where his body was later carried by the wind and current anyway. A whale traveling somewhere completely different cannot end up by chance exactly where the dead whale was found.

What we can say: The only explanation that fits everything—the recovery location, the date, the currents, and the depth profile of the sea—is that Timmy died shortly after his release on May 2. His carcass was then carried to Anholt by the wind and current in 12 days. The rescue operation failed—probably before the boats had left the horizon.

Why is this important?

Because thousands of people had real emotions. They worried, hoped, and cried—for a whale they had never met. those feelings were real. What they were told was not.

The initiative publicly communicated data that, according to the calculations of this study, could not physically exist. No raw data was published. No uncertainty was admitted. Instead, a picture was painted: Timmy is alive, Timmy is diving, Timmy is free.

Physics clearly reveals what actually happened: Timmy most likely died shortly after his release. In twelve days, quietly and without any activity, his body traveled the path to Anholt shown to it by the wind and current.

Whether there is intent behind the false data is a question this study cannot and does not want to answer. What it can answer: The data cannot be correct. And whoever communicated it knew it. Data that cannot physically exist cannot be transmitted by mistake.

This is no small matter.

Science cannot take away grief. But it can show what really happened.

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